Security and Cryptography
Item 2 - Cryptography Portfolio (CW)

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Code	M22359/M33122 Security and Cryptography
Purpose	This document specifies the lab exercises to be completed for the cryptography component of the module; your portfolio of answers to the lab exercises are to be submitted as an assignment worth 10% of the overall module mark.
Version	1.3 — Correction made to Exercise 4c *The two ciphertexts to compare are 4a and 4b. (corrected from 5a and 5b in earlier versions)* Supersedes: 1.2 — Correction made to Exercise 3d. *The first two letters of the plaintext are MA (corrected from ST in earlier versions).* 1.1 — Corrections made to Exercise 5. *The pdfs to be hashed in Exercise 5b and 5c are flamingo.pdf and hatter.pdf (rather than shattered-1.pdf and shattered-2.pdf used in the examples). The messages to be verified against the given signature in Exercise 5d (and the public key used to perform the verification) have also been revised in this version.* 1.0 — Initial Release

Schedule & Deliverables

Item	Value	Format	Deadline
Item 2 - Portfolio (CW)	10%	PDF	2024-03-13 16:00 **GMT**

Notes

- Submit your answers to the following exercises in one single PDF file to the submission dropbox on moodle.
- Use the accompanying examples to help you complete the following exercises; together, the examples and exercises help develop a technical understanding of a broad range of ciphers.
- This assignment rewards you for completing the lab exercises.
- Each week of content delivery / contact time supports one lab example/exercise.
- Remember: we’re here to support you as you build, develop and refine your knowledge and skills and to help you rise to the challenges set.
- This is individual work. You *will* learn extensively from each other, but any work you submit must be your own.

SECRYPT - Security and Cryptography - Exercises

Use the accompanying examples to help you complete the following exercises; together, the examples and exercises help develop a technical understanding of a broad range of ciphers.

The ciphertexts to be deciphered within the examples and exercises (e.g., `EXAMPLE_1` and `CIPHERTEXT_1B`) can all be found in `ciphertexts.py`

A number of the exercises require the encryption of varying amount of characters of a string called `PLAINTEXT` that is generated using the function `generate_plaintext(num)` in `secrypt.py`

```
>>> generate_plaintext(num)
```

This function takes an integer as input and returns a string that substitutes each number for a readable text version of the number in upper case and repeats this string twice.

```
input:      0     1     2     3     4     5     6     7     8     9
output:  ZERO   ONE   TWO THREE  FOUR  FIVE   SIX SEVEN EIGHT  NINE
```

To generate your value for `PLAINTEXT` enter the last three digits of your student number.

So, for example:
student number ending 056 generates a plaintext of ZEROFIVESIXZEROFIVESIX,
student number ending 119 generates a plaintext of ONEONENINEONEONENINE,
student number ending 123 generates a plaintext of ONETWOTHREEONETWOTHREE.

Here's an example of using the python function to generate the plaintext:

```
>>> PLAINTEXT = generate_plaintext('123')
>>> print(PLAINTEXT)
ONETWOTHREEONETWOTHREE
```

Make a copy of the Item 2 Template file and populate it with your student number and answers to each of the following exercises; do not change the formatting.

Print to PDF and submit this single PDF file to the submission dropbox on moodle.

Exercise 1 - Substitution Ciphers

a. Encrypt the first eight characters of `PLAINTEXT` with the following substitution:

```
input:  A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
output: A C E G I K M O Q S U W Y B D F H J L N P R T V X Z
```

[5 marks]

`CIPHERTEXT_1B` was encrypted using a monoalphabetic substitution cipher with an unknown key/mapping. Use letter frequency analysis and/or a dictionary attack to decipher.

b. Decipher `CIPHERTEXT_1B` [5 marks]

`CIPHERTEXT_1C` was encrypted using a monoalphabetic substitution cipher with an unknown key/mapping. Use letter frequency analysis and/or a dictionary attack to decipher.

c. Decipher `CIPHERTEXT_1C` [5 marks]

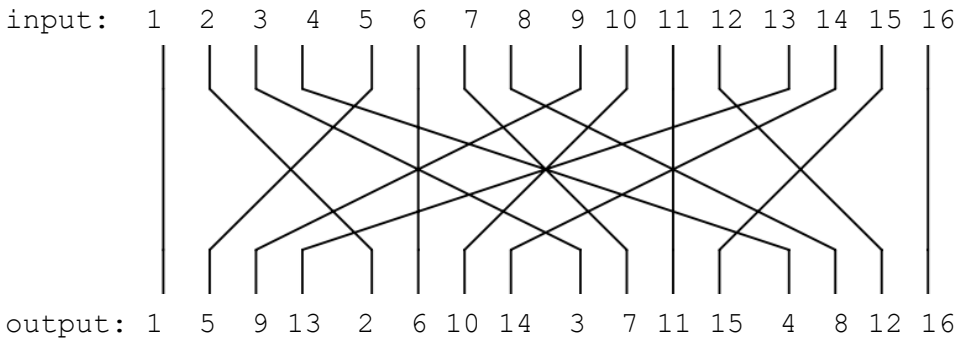
`CIPHERTEXT_1D` was encrypted using the Vigenere substitution cipher with an unknown key. Use the index of coincidence and letter frequency analysis to decipher.

d. Decipher `CIPHERTEXT_1D` [5 marks]

Exercise 2 - Transposition and XOR Ciphers

a. Encrypt the first sixteen characters of `PLAINTEXT` using the Columnar Transposition Cipher using a 4x4 grid with the columns transposed using an the keyword 'HARE' [5 marks]

b. Encrypt the ASCII binary encoding of the first two characters of `PLAINTEXT` using the following Permutation Box. [5 marks]



`CIPHERTEXT_2C` and `CIPHERTEXT_2D` have been encrypted using the XOR Cipher with the same key. Recall from `EXAMPLE_5` how you may use the properties of XOR to begin cracking the two plaintexts and the key. Recall also that when two ciphertexts are encrypted with the same key are XORed together, you get the two plaintexts XORed together.

c. Decipher `CIPHERTEXT_2C` [5 marks]

d. Decipher `CIPHERTEXT_2D` [5 marks]

Exercise 3 - Substitution Permutation Networks

a. Encrypt the first two letters of `PLAINTEXT`, using just one round of the Heys cipher where:

```
k  = 'A71C'
```

[5 marks]

b. Encrypt the first two letters of `PLAINTEXT`, using all four rounds of the Heys cipher where:

```
k  = 'A71CEA71CEA71CEA71CE'
```

[5 marks]

`CIPHERTEXT_3C` was encrypted using all four rounds of the Heys cipher where:

```
k  = 'CA75EA75BA75EA75CA75'
```

The same key was used to encrypt each block (i.e., the plaintext was encrypted using ECB mode).

c. Decrypt `CIPHERTEXT_3C` [5 marks]

`CIPHERTEXT_3D` was encrypted using just the first round of the Heys cipher using an unknown key. However, it is known that the first two letters of plaintext are 'MA', and that the same key was used to encrypt each block (i.e., the plaintext was encrypted using ECB mode).

d. Decipher `CIPHERTEXT_3D` [5 marks]

Exercise 4 - Block Cipher Modes of Operation

a. Encrypt `pt` using just the first round of the Heys cipher in ECB mode with key `k` where:

```
pt = PLAINTEXT[:4]*2
k  = 'A71C'
```

[5 marks]

b. Encrypt `pt` using just the first round of the Heys cipher in CBC mode with key `k` and initialisation vector `iv`, where:

```
pt = PLAINTEXT[:4]*2
k  = 'A71C'
iv = 'D0D0'
```

[5 marks]

c. What do you observe when comparing the two ciphertexts generated in answer to Exercise 4a and Exercise 4b? [5 marks]

`CIPHERTEXT_4D` was encrypted using all four rounds of the Heys Cipher using CBC mode with key `k` and initialisation vector `iv`, where:

```
k  = 'A71CEA71CEA71CEA71CE'
iv = 'D0D0'
```

d. Decrypt `CIPHERTEXT_4D` [5 marks]

Exercise 5 - Asymmetric Cryptography

a. Encrypt your student number (as an integer) using the following public key:
public key = (e,n) = (65537, 38083040388480806513)
[5 marks]

b. Generate a SHA1 hash of `hatter.pdf` and of `flamingo.pdf`
Compare the two hashes, what do you observe? [5 marks]

c. Generate a SHA256 hash of `hatter.pdf` and of `flamingo.pdf`
Compare the two hashes, what do you observe? [5 marks]

`SIGNATURE_5D` in `ciphertexts.py` was generated by using RSA with the private key paired with the following public key:
public key = (e,n) = (65537,
783138054801717533891525831983391189393125333089350836587983691486933484213232146961846790397734912297337647336825662827001219503683746071998839246345093 3)

d. Verify which of the following three messages is the genuine message

```
MESSAGE_1 = "Would you tell me, please, which way I ought to go from here? That depends a good deal on where you want to get to".encode('utf8')

MESSAGE_2 = "Its no use going back to yesterday, because I was a different person then.".encode('utf8')

MESSAGE_3 = "Begin at the beginning, the King said, very gravely, and go on till you come to the end: then stop.".encode('utf8')
```

[5 marks]